Semantic Propagation and Cross Modality Query Expansion in Content-Based Image Retrieval[1]

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ABSTRACT

Because of the fundamental problem unsolved on the mapping between low-level features and high-level semantics in Content-based image retrieval (CBIR), many practical systems still rely on text retrieval technologies by manual labeled keywords whereas such approach is huge labor intensive and thus unpractical for large image database. In this paper, we present the idea of semantic propagation based on learning-enhanced relevance feedback. That is, in general, there are two different modes of user interactions involved in typical retrieval systems: using keyword to represent semantic contents of the desired images, or query by examples based on low-level features. In most image retrieval systems, these two modes of interaction are mutually exclusive. We argue that combining these two approaches and allow them to benefit from each other yields a great deal of advantages in terms of both retrieval accuracy and ease of use. That is, at the very beginning only a few images in a database have been manual labeled with keywords and the retrieval is performed mainly based on low-level features. While the user is interacting with the system by providing feedbacks in a query session, a progressive learning process is activated to propagate the keyword annotation from the labeled images to un-labeled images so that more and more images are implicitly labeled by keywords at certain probabilities. In this way, more and more images are implicitly labeled by keywords by the semantic propagation process. Thus, such process will improve the

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retrieval performance in future, either query by image examples or by keywords. The significantly improved retrieval results show the effectiveness of our framework.

KEYWORDS: Content-based image retrieval (CBIR), Semantic Propagation, relevance feedback

1. INTRODUCTION

Having been actively researched on in the last decade [1][13][21], content-based image retrieval (CBIR) tries to solve the problems in automated indexing and retrieval on image database. Such approaches work with descriptions based on properties that are inherent in the images, such as color and texture. However, despite many research efforts, the retrieval accuracy of today's CBIR algorithms is still limited because the big gap between low-level features and semantic image contents. For different queries, different types of features have different significance; an issue is how to derive a weighting scheme to balance the relative importance of different feature types; there is no universal formula for all queries. The relevance feedback (RF) technique can be used to shorten or bridge the gap [3][14][15][19][26][27][28].

Relevance feedback, originally developed for Information Retrieval (IR)[18], is a supervised learning and active technique used to improve the effectiveness of information retrieval systems. The main idea of RF is using positive and negative examples provided by the user to improve system's performance. For a given query, the system first retrieves a list of ranked images according to a predefined similarity metrics, which often defined by the distance between query vector and feature vectors of images in a database. Then, the user select a set of positive and/or negative examples from the retrieved images, and the system will then refine the query and retrieves a new list of images. The key issue is how to incorporate positive and negative examples to refine the query and to adjust the similarity measure according to the feedback.

The original RF method can be illustrated by the Rocchio's formula [18] in Equation 1 in which the vector model [1][22][23] is used for IR. For a set of relevant documents D'_R and non-relevant documents D'_N given by the user, the new query is moved towards good example points and away from bad example ones.

$$Q' = \alpha Q + \beta\left(\frac{1}{N_{R'}}\sum_{i \in D'_R} D_i\right) - \gamma\left(\frac{1}{N_{N'}}\sum_{i \in D'_N} D_i\right) \quad (1)$$

where α, β, and γ are suitable constants; $N_{R'}$ and $N_{N'}$ are the number of documents in D'_R and D'_N respectively. This technique is also implemented in many CBIR systems[10][14]. Experiments show that the retrieval performance can be improved considerably by using this approach. General speaking, previous RF methods can be classified into two approaches: "weighing approach" and "probability approach". Most of existing works in CBIR use the former approach.

The weighting method [10][19][21] associates larger weights with more important dimensions and small weights with unimportant ones. For example, the paper in [21] generalizes a relevance feedback framework from the low-level feature based relevance feedback methods in existence. An ideal query vector for each feature *i* is described by the weighted sum of all positive feedback images as follows.

$$q_i^{T*} = \frac{\pi^T Y_i}{\sum_{n=1}^{N} \pi_n} \qquad 2$$

where Y_i is the $N \times K_i$ training sample matrix for feature *i* and is obtained by stacking the *N* positive feedback training vectors X_i^+ into a matrix. K_i is the length of the ith feature vector. The *N* element vector $\pi=[\pi_1,...\pi_N]$ represents the degree of relevance (to the query) of each of the *N* input positive feedback images, which can be determined by the user at each feedback iteration. The system then uses q_i^* as the optimal query to evaluate the relevance of the images in database. This strategy is widely used by many other image retrieval and relevance feedback systems.

Bayesian estimation methods have been used in the latter approach. Cox et al. [3], Vasconcelos and Lippman [28], Meilhac and Nastar [15] used Bayesian learning to incorporate user's feedback to update the probability distribution of all the images in the database. They considered the feedback examples as a sequence of independent query and try to minimize the retrieval error by Bayesian rules. That is, given a sequence of queries, try to minimize probability of error retrieval criteria as following equation.

$$\begin{aligned} g(x) &= \arg\max_i P(y=i \mid x_1,...,x_t) \\ &= \arg\max_i \{P(x_t \mid y=i) P(y=i \mid x_1,...,x_{t-1})\} \end{aligned} \qquad 3$$

where $\{x_1,...x_t\}$ is a sequence of queries (feedback examples), $P(y = i \mid x_1,...,x_t)$ is a prior belief about the ability of the ith image class to explain the query.

Thus, many practical CBIR systems are built based on the human labeling where each image in the database is manually labeled by a set of keywords and the retrieval problems are transferred to the text-based field. However, the process of manual indexing suffers from two major problems: huge labor-intensive labeling process and different subjectivity of human perceptions. These two issues limit the applications of such technology in large image database.

In general, there are two different modes of user interactions involved in typical retrieval systems: using keyword to represent semantic contents of the desired images, or query by examples based on low-level features. In most image retrieval systems, these two modes of interaction are mutually exclusive. We argue that combining these two approaches and allow them to benefit from each other yields a great deal of advantages in terms of both retrieval accuracy and ease of use. Based on this idea, we implement the semantic propagation scheme to improve the retrieval performances. That is, we assume that only a few images in a database have been assigned keywords and all images have been indexed with low-level features. In a retrieval session either the query is initialed with a keyword or an example image, by relevance feedbacks provided by the users, the progressive learning processes are activated to propagate the keyword annotations from the labeled images to un-labeled ones based on low-level similarities. In this way, more and more images are implicitly labeled by keywords by the semantic propagation process. Thus, such process will further improve the retrieval performance in future, either query by image examples or by keywords.

This paper is organized as follows. In Section 2 we present the framework and unified similarity measure of the retrieval system '*MiAlbum*'. Section 3 introduces the proposed relevance feedback algorithm. Section 4 presents the propagation algorithm and query expansion process. Experimental results are shown in Section 5.

2. UNIFIED FRAMEWORK AND SIMILARITY MEASURE

Generally, two modes of user interactions involved in current CBIR systems. One is query by keywords, that is, the user types in a list of keywords representing the semantic contents of the desired images and submits to the retrieval system to find his/her desired images. The other method is query by example. In this case, the user provides example image(s) as the input and the retrieval system will try to retrieve other similar images. In most image retrieval systems, these two modes of interaction are mutually exclusive. We argue that combining these two approaches and allow them to benefit from each other yields a great deal of advantage in terms of both retrieval accuracy and ease of use of the system.

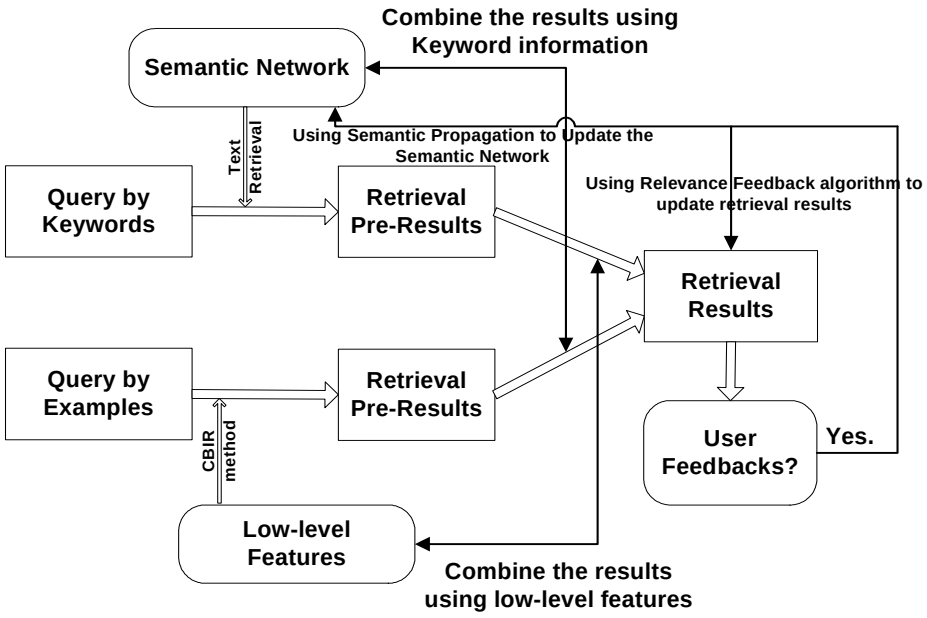


Figure 1. System architecture of *MiAlbum*: Integrated Relevance Feedback and Query Expansion.

Figure 1 demonstrates the proposed retrieval framework. It is a CBIR system supporting both query by keyword and query by example methods. Besides relevance feedback, our system also supports cross-modality query expansion and the strategy of the semantic propagation. We will introduce these algorithms in detail at the latter sessions. From the figure we can see that once a query is submitted in the form of keyword, the keyword retrieval method can provide its result list. We call such list as Retrieval Pre-results. Then the query is expanded by features of the images of

this list and such low-level features are considered as the positive examples. Then a hidden low-level feature based feedback process will try to refine the retrieval results and show the final result list to the end user. For the query by example, similar procedure takes effect to extend the retrieval from feature space to semantic space. This is the query expansion process. Different from the previous expansion schemes in IR, our expansion is performed between low-level feature space and high-level semantic space. It can utilize the information as much as possible to improve the retrieval performance.



Figure 2. Retrieval and feedback interface of *MiAlbum*. Example image is shown on the left of the bottom.

Beside the hidden feedback, user can refine his/her retrieval by relevance feedback method. The main user interface of *'MiAlbum'* is shown in Figure 2. It has a simple interface to search images by keywords or examples. User can drag and drop the sample image to bottom-left image view to query by example or just type the keyword(s) in text box to search by keyword. Results are returned as a ranked list show in the right image view. The ranking sequence is from left to right, from top to

bottom. During the retrieval process user can provide relevance feedback by click the 'Yes' or 'No' sign below the results. As we can there is an Option Dialog in the main interface. User can select the features and methods, as well as the selection between original features space and PCA feature space.

User' feedback interactions not only improve his/her current retrieval performance. Moreover, such feedback will also update the semantic network (describe later) by semantic propagation method so that system performance can be iteratively improved over time. That is to say, the framework can memorize user's feedback information to benefit latter retrieval sessions. General to say, the goal of user's feedback is to generate a set of query that is guaranteed to contain the user's intended search results. The query can then be narrowed down by including more feedback images through the relevance feedback cycle. Moreover, user's feedback will update the semantic network according to the routine described in the latter section. It will improve the initial retrieval performance for the latter usage.

Both the hidden feedback and user feedback processes have to integrate the low-level feature based feedback and high-level semantic feedback together. We define a unified metric function *G* to measure the relevance between any image *j* and query $Q^{'}$ within the image database in terms of both semantic and low-level feature content, where $Q^{'}$ includes the original query and users' feedback information.

$$G(j,Q^{'}) = \alpha \cdot sim_k(j,Q^{'}) + (1-\alpha) \cdot sim_f(j,Q^{'}) \tag{4}$$

In the upper equation, $\alpha \in [0,1]$ is the weight of the semantic relevance in the overall similarity measurement which can be specified by users. The larger α is, the more important the semantic relevance will play in the overall similarity measurement. As the *MiAlbum* interface shown in Figure 2, it can be set manually by tuning the box position of the slide-bar in bottom-left corner. $sim_f(j,Q^{'})$ is the low-level feature similarity between the images *j* and $Q^{'}$. It can be calculated out by algorithm described in section 3.1. $sim_k(j,Q^{'})$ is the distance in semantic space. It is calculated by the standard Rocchio's formula shown in section 3.2. Both distances have been normalized to the same scope.

The general structure of our framework has been introduced in the previous part. In the following sections, we will describe the algorithm in each component in detail.

3. PROGRESSIVE LEANING IN RELEVANCE FEEDBACK

In this section, we will introduce the proposed relevance feedback algorithm. Because our framework can support both query by example and keyword methods, the feedback algorithm include two components, one is in low-level feature space, and the other is in semantic space.

3.1. Feedback in low-level feature space

Our idea about relevance feedback in low-level space is the following: All positive examples and query image in one retrieval session can be considered as a semantic class in which all images contain the same object(s) or represent similar semantic meaning(s). We assume that the low-level feature distribution of such semantic class is the Gaussian distribution.

Suppose that *m* types of features, i.e., an image is represented by $\vec{x} = [\vec{x_1}, ..., \vec{x_m}]$, where $\vec{x_i}$ is the feature vector in the K_i-dimensional feature space $\mathbf{S}_i$ for type *i* features. Since we assume that all positive examples in one retrieval session belong to the same semantic class and their corresponding feature sets following the Gaussian distribution, $\vec{x_i} \sim N(\varepsilon_i, \Sigma_i)$, where Σ_i is the $K_i \times K_i$ dimension covariance matrix and ε_i is the n_i dimensional mean vector. Let *n* be the number of positive feedbacks for image *P*. We use a diagonal matrix $diag\{\sigma_i^2\}$, where $\sigma_i(m) = \Sigma_i(m, m)$, to represent the inter-feature correlation, for simplicity. Consider vector *x* obeys Gaussian distribution; then, the probability density function of *x* is:

$$p(x) = \frac{1}{(2\pi)^{d/2} |\Sigma_i|^{\frac{1}{2}}} e^{-\frac{1}{2}(x-\varepsilon_i)^T \Sigma_i^{-1} (x-\varepsilon_i)} \quad (5)$$

There are two parameters in this function: $\varepsilon_i, \Sigma i$. They need to be updated when more positive examples are provided by the user through relevance feedback. Such updating can be considered as the combination of two Gaussian classes, one is the original class; the other is constructed by the new

positive examples. Denote the current set of positive examples by *U*. The updating can be performed as follows

$$\sigma'_i{}^2 = n\sigma_i^2 + \frac{n|U|\varepsilon_i^2 - 2n\varepsilon_i \sum_{u\in U} u}{n+|U|} + \sum_{u\in U} u^2 - \frac{(\sum_{u\in U} u)^2}{n+|U|} \tag{6}$$

$$\varepsilon'_i = \frac{n\times\varepsilon_i + \sum_{u\in U} u}{n_k + |U|} \tag{7}$$

$$n' = n + |U| \tag{8}$$

Thus, the parameters for the semantic Gaussian class can be estimated using the feature vectors of all the positive examples. The log posterior probability that the feature vector x belongs to the subjective class c_i implied in the positive examples is estimated using the following Bayesian formulation

$$\begin{aligned} g_i(x) &= \ln P(c_i \mid x) \propto \ln\, p(x \mid c_i) + \ln P(c_i) \\ &= -\frac{1}{2}(x-\varepsilon_i)^T \Sigma_i^{-1}(x-\varepsilon_i) + \ln P(c_i) + const \end{aligned} \tag{9}$$

where *const* are fixed in one RF iteration. This formula is for one feature type. Assuming that different feature types are independent of each other, the log posterior is calculated as the sum of individual $g_i(x)$'s.

The following describes how positive feedbacks are performed:

Initialization of System

1. Feature Normalization: This allows equal emphasis on all the feature components. For $\vec{x}_i$ the normalized vector is $\vec{x}'_i = \lfloor x_{i1},...,x_{in_i} \rfloor$, where $x'_{i_m} = \frac{x''_{i_m} - \varepsilon(x''_{i_m})}{3\sigma(x_{i_m})}$ and $x''_{i_m} = \frac{x''_{i_m} - \min(X_{i_m})}{\max(X_{i_m}) - \min(X_{i_m})}$. If $x_{i\,m}$ satisfies the Gaussian distribution, it is easy to prove that the probability of $x'_{i\,m}$ being in the range of [-1,1] is 99%.

2. Initialization: Initialize $\sigma_i = \mathbf{I}$ (identity matrix) and $\varepsilon_i = \vec{x}_i$, $n = 1$.

Retrieval and Feedback

1. Update the retrieval parameters σ_i , ε_i , and n according to Equation (6)-(8) using the information provided by the current set of positive examples U.

2. Distance Calculation: For each image *Pj*, $P_j \in D$, its similarity to current query session $sim_f(j,Q')$ is calculated using Equation 9 in the retrieval after the feedback, where $sim_f(j,Q') = g_{Q'}(P_j)$, That is, the similarity of each image in the database to be refined query is determined by Equation 9 based on the positive examples.

3. Combining the similarities in feature and semantic spaces and sorting. Then provide the new ranking list to the user.

Our feedback algorithm is incremental. This has the following advantages: Not only the retrieval performance could be improved for the current user, but also the improvements also help the subsequent users. The parameters of the Gaussian semantic class corresponding to each query will be refined by its positive examples and could be stored for the latter usage by others. The updating process is an online process, so the computation complexity is quite important. However, such updating process only needs taking care of the current positive examples. So the computation cost could be negligent as compared with the whole retrieval process.

3.2. Feedback in semantic (keyword) space

The proposed *MiAlbum* relevance feedback framework consists of a semantic network from an image database that links images to semantic annotations, a similarity measure that integrating both semantic features and image features, and a machine learning algorithm to iteratively update the semantic network and to improve the system's performance over time. It is a two-layered structure. Top layer is represented by a set of keywords having links to the images in the database. Weights are assigned to each individual link, as shown pictorially in Figure 3. The degree of relevance of the keywords to the associated images' semantic content is represented as the weight on each link. This

layer is what we need in keyword relevance feedback and will be updated during the semantic propagation. Bottom layer is a keyword thesaurus to construct the connection between different keywords. Wordnet is used here as the query expansion tool to help keyword search in image retrieval.

Semantic based relevance feedback is performed in keyword vector space. Each dimension is represented by a specific keyword. For each image, there are corresponding text descriptors composing by a text feature vector. The text features can be got by manual labeling or if the image is downloaded from website, more keywords can be assigned according to its URL, filename, ALT text, hyperlinks and surrounding text. Of course if no keyword information to the image, the corresponding feature vector is set to null. We will use the semantic propagation method to annotate such image during user's feedback iteration. This will be discussed later.

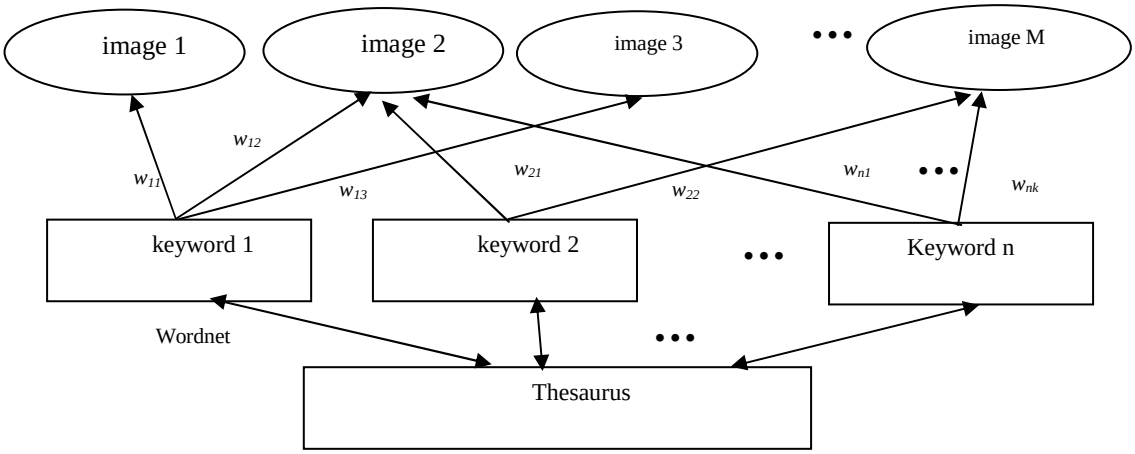


Figure 3. Semantic network of the image database.

Semantic based relevance feedback can be performed relatively easily compared to its low-level feature counterpart. The structure of proposed semantic network is shown in Figure 3. As we can see from the figure, there are links to associate the images in DB with the keywords in Thesaurus dictionary. The corresponding weight of such association link describes how close between the specific image and keyword. The initial value of such weight w_{ij} is calculated by the TF*IDF method [22].

$$w_{ij} = TF * IDF = n_{ij} * \log(\frac{N}{n_i}) \tag{10}$$

where N is the number of images having non-null keyword feature vectors of the whole image database, n_i is the number of the images having the keyword j, $n_{ij} = \frac{a_{ij}}{\sum_i a_{ij}}$ and a_{ij} is annotation times of keyword i to image j, or if the image is from web and indexed by its hyperlink and page information, a_{ij} can be considered as the appearance times of keyword i to image j. Then we can get the keyword vector K_i for image i. The basic idea of semantic feedback can be concluded as the following simplified Rocchio's formula.

$$Q' = Q + \frac{1}{N_{R'}} \sum_{i \in D'_R} K_i - \frac{1}{N_{N'}} \sum_{i \in D'_N} K_i \tag{11}$$

Then we can get the similarity between any keyword vector and Q' according as follows:

$$sim_k(j, Q') = \frac{|Q' \bullet K_j|}{|Q'| \cdot |K_j|} \tag{12}$$

The feedback process in semantic space can be described as below:

1. User input keyword query *Q*
2. Using cosine similarity to calculate the distance between query and each image of the database
3. Sorting the distance and return the result list to user
4. If no feedbacks, end this retrieval session
5. or collect the positive and negative feedback examples
6. Modify the *Q* according to Equation (11).
7. Return to step 2.

4. PROBABLISTIC PROPAGATION SCHEME

While image database systems built based on the human labeled keywords tends to perform better compared to CBIR systems, the limitations are: labor-intensive labeling process and subjectivity of human perception. To address this issue, while improving the performance of CBIR approaches, we have developed a framework that performs relevance feedback and query refinement on both the images' semantic contents represented by keywords and the low-level feature vectors. In

other words, semantic and low-level feature based relevance feedback are seamlessly integrated. Only when the semantic information is not available, our method is reduced to one of the previously described low-level feedback approaches as a special case.

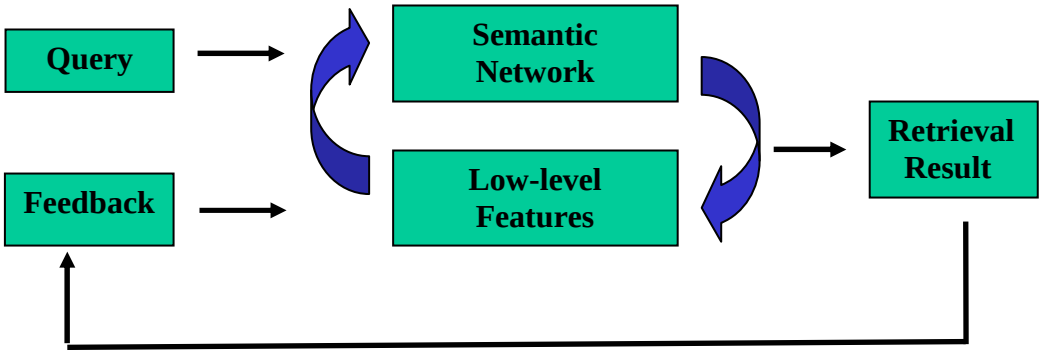


Figure 4. Propagation Framework of *MiAlbum*

We assume that only a few images in a database have been manual labeled with keywords and the retrieval is performed mainly based on low-level features. The initial keywords annotation can be from web through the crawler when the images are from the Web, or labeled by humans. While the user is interacting with the system by providing feedbacks in a query session, a progressive learning process is activated to propagate the keyword annotation from the labeled images to un-labeled images so that more and more images are implicitly labeled by keywords. In this way, the semantic network is updated in which the keywords with a majority of user consensus will emerge as the dominant representation of the semantic content of their associated images. As more queries are inputted into the system, the system is able to expand its vocabulary. Also, through the propagation process, the keywords that represent the actual semantic content of each image will receive a large weight.

There are two major barriers in keyword propagation: which images and which keyword(s) should be propagated during a query session.

To answer the first question, a probability model, based on Bayesian learning, is proposed. We assume that, (1) all positive examples in one retrieval session belong to the same semantic class with common semantic object(s) or meaning(s); and (2) the features from the same semantic class follows the Gaussian or Mixture Gaussian distributions. Therefore, all positive feedbacks in a query session

are used to calculate and update the parameters of the corresponding semantic Gaussian classes. Then, the probability of each image in the database belonging to such semantic class is calculated. The common keywords in positive examples are propagated to the images with very high probability belonging to this class.

As we can see, the propagation framework uses the same procedure as the feedback algorithm in low-level features. The only different is, for low-level feature feedbacks, the calculated probability is used for the ranking and here it is used to determine whether or not it should be in the propagation candidate list.

Same as the feedback algorithm $p(c_i)$ can be calculated according to Equation 9. Then we can obtain the propagation candidate set S as follows:

$$S = \{c_1, ..., c_k\}, \text{ where } p(c_i) > \psi \quad (13)$$

ψ is a constant threshold which can be estimated by the training process. The weight associate with the propagated keyword *i* and the image *m* $w_{im} = p(c_m)$.

More complex distribution model, for example Mixture Gaussian, could be used in this propagation framework. However, because of user's very few feedback examples in practical use, the complex models have more parameters to estimate and it will leads into much more parameter estimation errors.

Also, to determine which keyword(s) should be propagated when an image is associated with multiple keywords, a region-based approach is evolved. Unlabelled images to be propagated are firstly segmented into regions. By analyzing the feature distribution of the segmented regions, a probability association between each segmented regions and annotated keywords is set up for labeled images by region-based relevance feedback approach. Then, each keyword of labeled image was assigned to one or several regions of the image with certain probabilities. The region-based feedback framework we used is described at [11]. It is also a probabilistic feedback approach. The segmentation method used is JSEG [4]. It is a color based segmentation algorithm. We are working on this project and still at very early stage. So, at current work, all common keywords will be propagated to the candidate list.

With the propagation method described before, more images will have their keyword information by user's feedbacks. That will help the retrieval system in two aspects. First, more images can support keyword search. Second, the user's feedback information has been stored in such propagation scheme so that it will be helpful for future retrieval requests.

5. EXPERIMENTAL RESULTS



(a) Results without relevance feedback (b) Results after relevance feedback (3 positive examples)



(c) Propagation candidate besides positive examples

Figure 5: One demo Retrieval Session.

The image set we used is the Corel Image Gallery. 10,000 images of 79 semantic categories are selected to calculate the performance statistics. 200 random selected images compose the standard query set. Whether a retrieved image is correct or incorrect is judged according to the ground truth class. Three types of color features and three types of texture features are used in our system, as shown in Table 1.

Table 1: Low level features used in '***MiAlbum***'.

Features	Dimension
Color histogram in HSV space with quantization 256	256
First and second color moments in Lab space	9
Color coherence vector in LUV space with quantization 64	128
Tamura coarseness histogram	10
Tamura directionality	8
Pyramid wavelet texture feature	24

Feedback process is running as follows: Given a query example from the test set, one different test image of the same category as the query is used in each round of feedback iteration as the positive example for updating the Gaussian parameters. For negative feedback process the first two irrelevant images are assigned as negative examples. The accuracy is defined as:

$$Accuracy = \frac{\text{relevant images retrieved in top } N \text{ returns}}{N} \tag{14}$$

Several experiments are performed as follows. First, our Bayesian feedback scheme is compared with previous feedback approaches presented by Nuno [28] and Rui [19][21]. This comparison is done in the original feature space. Figure 6, Figure 7 and Figure 8 show that the accuracy of our Bayesian feedback method becomes higher than the other two methods after two RF iterations. This demonstrates that the incorporated Bayesian estimation with the Gaussian parameter-updating scheme can improve RF-based retrieval.

To demonstrate the performance of the semantic propagation, we design the following experiment. We simply annotate the 200 images in the query set by its category names. So, only one keyword is associated to one query picture and other images in database have no keyword annotations. During the test, each query image is used twice. Then we compare the retrieval

performance between feedback without propagation and feedback with propagation scheme. The results are shown in Figure 9 and Figure 10. As we can see from the figure, for feedback with propagation, the retrieval accuracy is much higher than the original one. That is because, without propagation scheme, the retrieval system has no 'memory'. Every time when user re-submits the same query, he/she has to start from the very low accuracy because system can not memorize the pervious users' feedback information. However, if system has propagation ability, latter user can get the initial retrieval results based on previous feedback iterations. In other words, system has the leaning ability and will be smarter and smarter with more and more users' interactions.

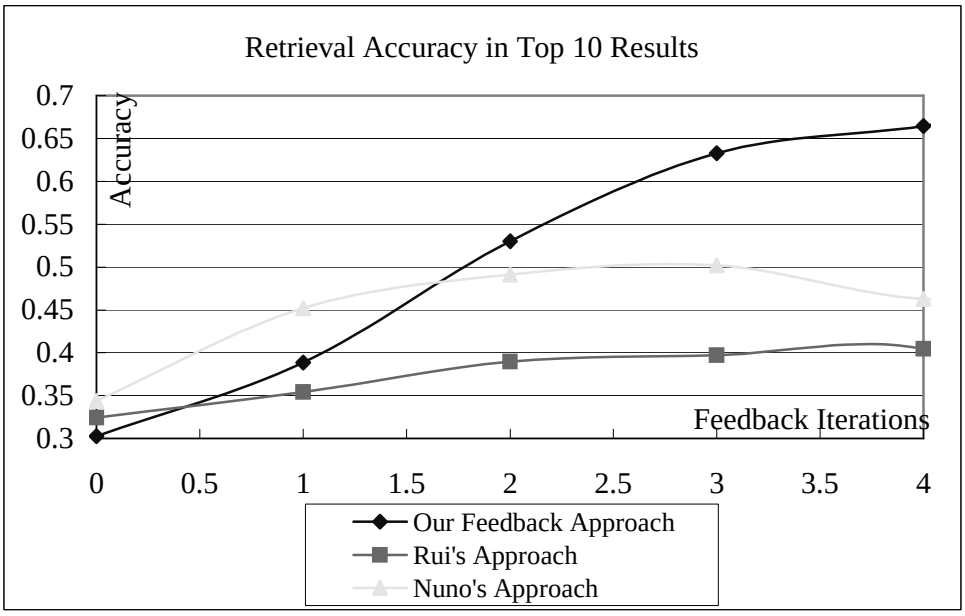


Figure 6: retrieval accuracy for top 10 results in original feature space.

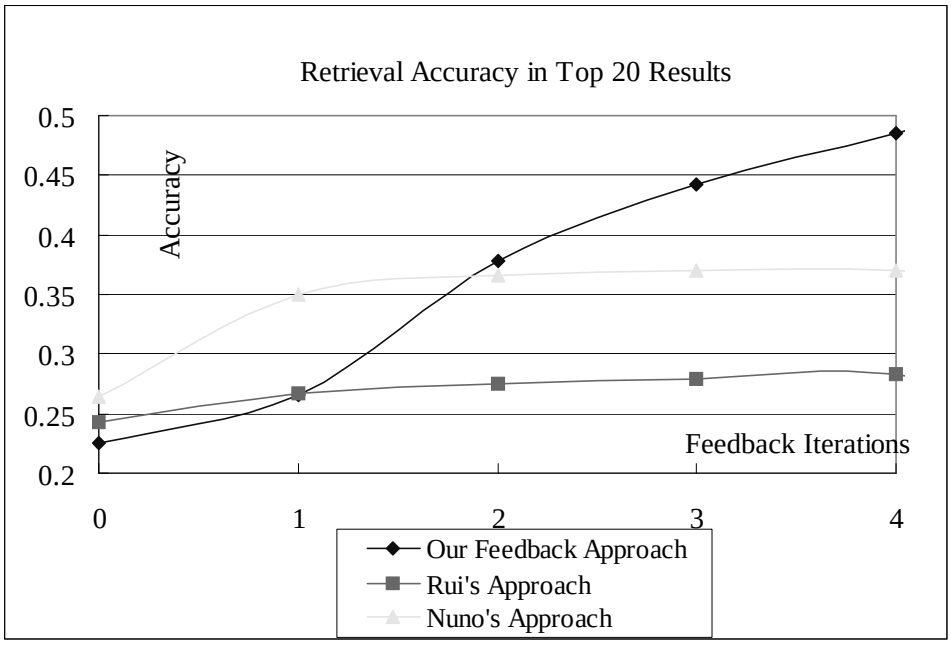


Figure 7: retrieval accuracy for top 20 results in original feature space.

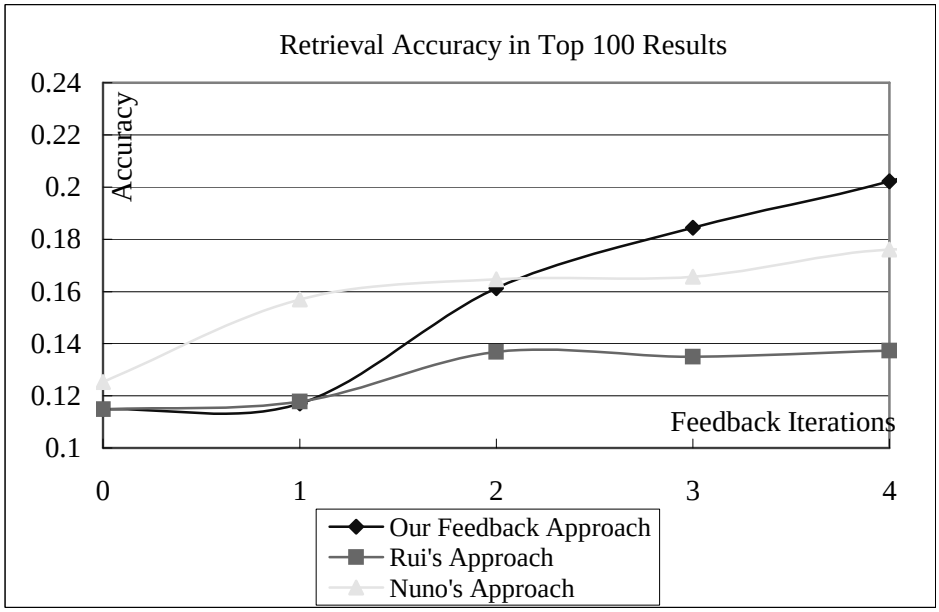


Figure 8: retrieval accuracy for top 100 results in original feature space.

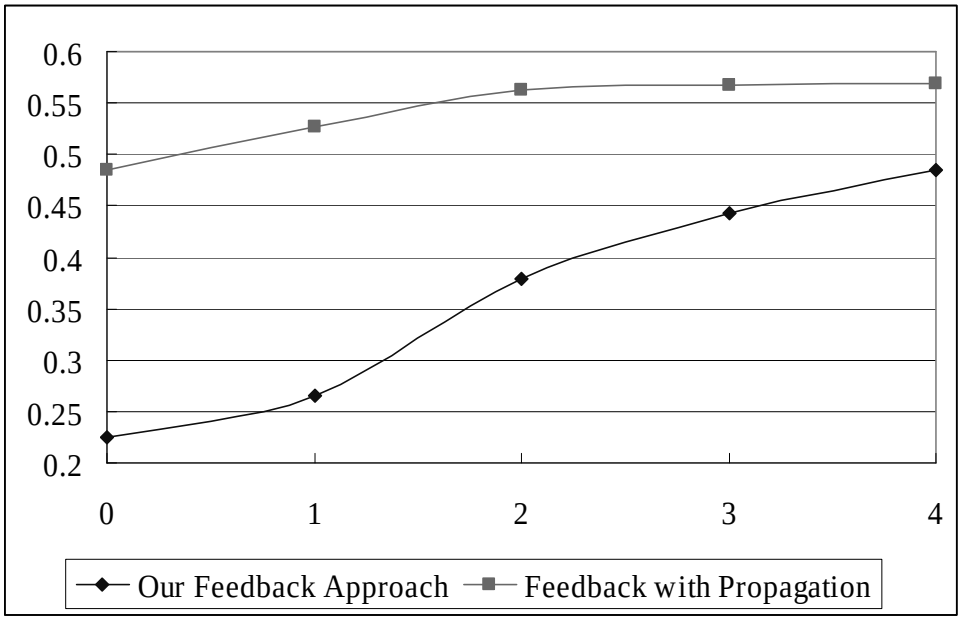


Figure 9: retrieval accuracy for top 20 results

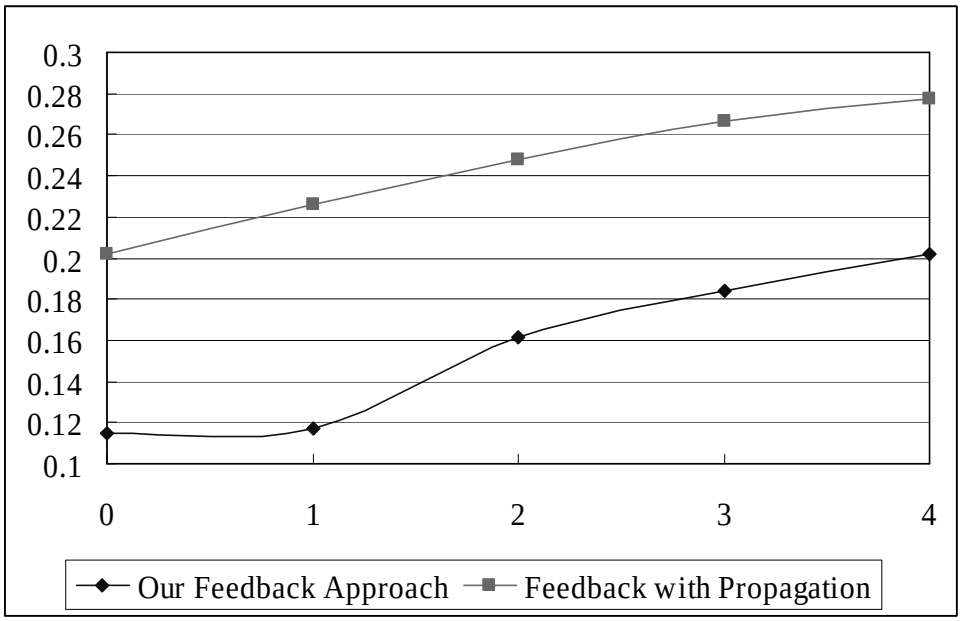


Figure 10: retrieval accuracy for top 100 results

6. CONCLUSION

In this paper, we present the framework design of our image retrieval system which is build based on relevance feedback and probabilistic keyword propagation. Our method utilizes both semantic and low-level feature properties of every feedback image in refine the retrieval, while in the

meantime, learning semantic annotations of each image. While the user is interacting with the system by providing feedbacks in a query session, a progressive learning process is activated to propagate the keyword annotation from the labeled images to un-labeled images so that more and more images are implicitly labeled by keywords at certain probabilities. In this way, more and more images are implicitly labeled by keywords by the semantic propagation process. Thus, such process will improve the retrieval performance in future, either query by image examples or by keywords. This new framework makes the image retrieval system to be superior over either the classical CBIR or text-based systems.

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